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10 Division of Polynomials

Defination :- If index of each term of an algebric expression in one variable is a whole number Then the expression is called polynomial.

Degree of polynamicals :- The greatest index of variable in the given polynomials is called the degree of the polynomial.

Practice Set 10.1

1. Divide. Write the quotient and the remainder.

① $21m^2 \div 7m$

$$\begin{array}{r} 3m \\ 7m\,\overline{)\,21m^2} \\ -21m^2 \\ \hline 00 \end{array}$$

Quotient = $3m$

Remainder = 00

③ $(-48p^4) \div (-9p^2)$

$$\begin{array}{r} \frac{16}{3}p^2 \\ 9p^2\,\overline{)\,48p^4} \\ -48p^4 \\ \hline 00 \end{array}$$

$$\boxed{\frac{\cancel{9}^{3} \times 16}{\cancel{3}} = 48}$$

Quotient = $\frac{16}{3}p^2$

Remainder = 0

④ $40m^5 \div 30m^2$

$$\begin{array}{r} \frac{4}{3}m^2 \\ 30m^2\,\overline{)\,40m^5} \\ -40m^5 \\ \hline 00 \end{array}$$

Quotient = $\frac{4}{3}m^2$

Remainder = 0

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(4) $40a^3 \div (-10a)$

$$\begin{array}{r} -4a^2 \\ -10a \overline{) 40a^3} \\ -40a^3 \\ \hline 000 \end{array}$$

$\frac{\cancel{40a^3}^{4}}{\cancel{-10a}_{1}}$

Quotient : $-4a^2$
Remender- 00

(5) $(5x^3 - 3x^2) \div x^2$

$\frac{5x^3 - 3x^2}{x^2}$

Quotient - $5x - 3$
Remainder- 0

$$\begin{array}{r} 5x - 3 \\ x^2 \overline{) 5x^3 - 3x^2} \\ - 5x^3 - 3x^2 \\ \hline 00000 \end{array}$$

(6) $(8p^3 - 4p^2) \div 2p^2$

$$\begin{array}{r} 4p - 2p \\ 2p^2 \overline{) 8p^3 - 4p^2} \\ - 8p^3 - 4p^2 \\ \hline 0000 \end{array}$$

Quotient - $4p - 2p$
Remainder- 0

(7) $(2y^3 + 4y^2 + 3) \div 2y^2$

$$\begin{array}{r} y + 2 \\ 2y^2 \overline{) 2y^3 + 4y^2 + 3} \\ - 2y^3 + 4y^2 \\ \hline 00003 \end{array}$$

Quotient = $y + 2$
Remainder = 3

(8) $(21x^4 - 14x^2 + 7x) \div 7x^3$

$$\begin{array}{r} 3x \\ 7x^3 \overline{) 21x^4 - 14x^2 + 7x} \\ -21x^4 \\ \hline 00 - 14x^2 + 7x \end{array}$$

$\frac{\cancel{21}^{3}x^4 - 14x^2 + 7x}{\cancel{7}_{1}x^3}$

Quotient - $3x$
Remainder -
$14x^2 + 7x$

(9) $(6x^5-4x^4+8x^3+2x^2) \div 2x^2$

$$
\begin{array}{r l}
 & 3x^3-2x^2+4x+x \\
2x^2 \,) & 6x^5-4x^4+8x^3+2x^2 \\
 & -6x^5 \\
 & 0\,0-4x^4 \\
 & -4x^4 \\
 & 0\,0+8x^3 \\
 & -8x^3 \\
 & 0\,0+2x^2 \\
 & -2x^2 \\
 & 0\,0
\end{array}
$$

Quotient $= 3x^3-2x^2+4x+x$

Remainder $= 0$

(10) $(25m^4-15m^3+10m+8) \div 5m^3$

$$
\begin{array}{r l}
 & 5m-3m+2m \\
5m^3 \,) & 25m^4-15m^3+10m+8 \\
 & -25m^4 \\
 & 0\;\;0-15m^3 \\
 & -15m^3 \\
 & 0\;\;0+10m \\
 & -10m \\
 & 0\,0+8
\end{array}
$$

Quotient $- 5m-3m+2m$

Remainder $-$ 8

Practice Set 10.2

1. Divide and ~~right~~ write the quotient and the remainder.

~~(1) $(y^2+10y+24) \div (y+4)$~~

~~$y+4 \overline{) y^2+10y+24}$~~

(4) $(2m^3+m^2+m+9) \div (2m-1)$

$$\begin{array}{r} m^2+m+1 \\ 2m-1 \overline{) 2m^3+m^2+m+9} \\ -2m^3+m^2 \\ \hline 0+2m^2+m \\ -2m^2+m \\ \hline 0+2m+9 \\ -2m+1 \\ \hline 0+10 \end{array}$$

$(2m-1) \times m^2 = 2m^3-m^2$

$(2m-1) \times m = 2m^2-m$

$(2m-1) \times 1 = 2m-1$

Quotient $= m^2+2+1$

Remainder $= 10$

(6) $(a^4-a^3+a^2-a+1) \div (a^3-1)$

$$\begin{array}{r} a-1 \\ a^3-2 \overline{) a^4-a^3+a^2-a+1} \\ -a^4 \quad -2a \\ \hline 0-a^3+a^2-a+1 \\ -a^3 \quad -2 \\ \hline 0+a^2-a-1 \end{array}$$

$(a^3-2) \times a = a^4-2a$

$(a^3-2) \times 1 = a^3-2$

Quotient $= a-1$

Remainder $= a^2-a-1$

(1) $(y^2+10$

$y+4$

(2) $(p^2$

$p+$

(3) $(a$

(4) $($

① $(y^2+10y+24) \div (y+4)$

$$\begin{array}{r} y+6 \\ y+4\overline{)\,y^2+10y+24} \\ -\,y^2+4y \\ \hline 0+6y+24 \\ -\,6y+24 \\ \hline 0\quad 0 \end{array}$$

$(y+4) \times y = y^2+4y$

$(y+4) \times 6 = 6y+24$

Quotient = $y+6$

Remainder = 0

② $(p^2+7p-5) \div (p+3)$

$$\begin{array}{r} p+4 \\ p+3\overline{)\,p^2+7p-5} \\ -\,p^2+3p \\ \hline 0\quad 4p-5 \\ 4p-12 \\ \hline 0\quad -17 \end{array}$$

$(p+3) \times p = p^2+3p$

$(p+3) \times 4 = 4p+12$

Quotient = $p+4$

Remainder = -17

③ $(3x+2x^2+4x^3) \div (x-4)$

$$\begin{array}{r} 4x^2 \\ x-4\overline{)\,4x^3+2x^2+3x} \\ -\,4x^3-16x^2 \\ \hline 0\quad +18x^2+3x \end{array}$$

$(x-4) \times 4x^2 = 4x^3-16x^2$

$(x-4) \times 18x = 18x^2 -$

④ $(2m^3+m^2+m+9) \div (2m-1)$

$$\begin{array}{r} m^2+m+1 \\ 2m-1\overline{)\,2m^3+m^2+m+9} \\ -\,2m^3-m^2 \\ \hline 0+2m^2+m+9 \\ -\,2m^2-m \\ \hline 0+2m+9 \\ -\,2m+1 \\ \hline 0+10 \end{array}$$

$(2m-1) \times m^2 = 2m^3-m^2$

$(2m-1) \times m = 2m^2-m$

$(2m-1) \times 1 = 2m-1$

Quotient = m^2+m+1

Remainder = 10

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(5) $(3x - 3x^2 - 12 + x^4 + x^3) \div (2 + x^2)$

$\rightarrow (x^4 + x^3 - 3x^2 + 3x - 12) \div (x^2 + 2)$

$$
\begin{array}{r}
x^2 + x - 5 \\
x^2+2 \,\overline{)\, x^4 + x^3 - 3x^2 + 3x - 12} \\
-x^4 \qquad + 2x^2 \qquad\qquad \\
\hline
0 + x^3 - 5x^2 + 3x - 12 \\
-x^3 \qquad + 2x \qquad \\
\hline
0 - 5x^2 + x - 12 \\
-5x^2 \qquad + 10 \\
\hline
0 + x - 2
\end{array}
$$

$(x^2+2) \times x^2 = x^4 + 2x^2$

$(x^2+2) \times x = x^3 + 2x$

$(x^2+2) \times 5 = 5x^2 + 10$

Quotient $= x^2 + x - 5$

Remainder $= x - 2$

(6) $(a^4 - a^3 + a^2 - a + 1) \div (a^3 - 2)$

$$
\begin{array}{r}
a - 1 \\
a^3-2 \,\overline{)\, a^4 - a^3 + a^2 - a + 1} \\
-a^4 \qquad\qquad - 2a \qquad \\
\hline
0 - a^3 + a^2 - a + 1 \\
-a^3 \qquad\qquad - 2 \\
\hline
0 + a^2 - a + 1
\end{array}
$$

$(a^3-2) \times a = a^4 - 2a$

$(a^3-2) \times 1 = a^3 - 2$

Quotient $= a - 1$

Remainder $= a^2 - a + 1$

(7) $(4x^4 - 5x^3 - 7x + 1) \div (4x - 1)$

$$
\begin{array}{r}
3 \qquad\qquad\qquad\qquad \\
4x-1 \,\overline{)\, 4x^4 - 5x^3 + 0x^2 - 7x + 1} \\
-4x^4 \qquad\qquad\qquad\qquad - 3 \\
\hline
0 \quad - 5x^3 + 0x^2 - 7x
\end{array}
$$

$(4x-1) \times 3 = 4x^4 - 3$